LABSHEET-10

File Handling

Q1. Create and Write to a File

Write a Java program to create a file named student.txt and write Student Name, Roll Number, Course, and Semester into it. Display a suitable message after the file is successfully created and data is written.

```
import java.io.File;
import java.io.FileWriter;
import java.io.IOException;

public class Main {
    public static void main(String[] args) {
        try {
            File file = new File("student.txt");
            if (file.createNewFile()) {
                System.out.println("File created: " + file.getName());
            } else {
                System.out.println("File already exists. It will be overwritten.");
            }

            FileWriter fw = new FileWriter(file);
            fw.write("Student Name : Aryan \n");
            fw.write("Roll Number  : CU25260022");
            fw.write("Course       : BCA\n");
            fw.write("Semester     : 2\n");
            fw.close();

            System.out.println("Data written successfully to student.txt");
        } catch (IOException e) {
            System.out.println("An error occurred: " + e.getMessage());
        }
    }
}
```

OUTPUT:-

```
File created: student.txt
Data written successfully to student.txt
```

Q2. Read Data from a File

Write a Java program to read the contents of student.txt created in Question 1 and display the complete contents on the console. If the file does not exist, display an appropriate error message.

```
import java.io.BufferedReader;
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.IOException;

public class Main {
    public static void main(String[] args) {
        try {
            BufferedReader br = new BufferedReader(new FileReader("student.txt"));
            String line;
            System.out.println("Contents of student.txt:");
            System.out.println("------------------------");
            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }
            br.close();
        } catch (FileNotFoundException e) {
            System.out.println("Error: student.txt does not exist.");
        } catch (IOException e) {
            System.out.println("Error while reading file: " + e.getMessage());
        }
    }
}
```

OUTPUT:-

```
Contents of student.txt:
------------------------
Student Name : Aryan
Roll Number  : CU25260022
Course       : BCA
Semester     : 2
```

Q3. Count Characters, Words and Lines

Create a text file named data.txt containing multiple lines of text. Write a Java program to read the file and calculate the number of lines, words, and characters. Display all three results.

data.txt (input file):

```
Java is a popular programming language.
File handling is an important topic in Java.
It allows us to read and write data.
```

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;

public class Main {
    public static void main(String[] args) {
        int lines = 0, words = 0, chars = 0;
        try {
            BufferedReader br = new BufferedReader(new FileReader("data.txt"));
            String line;
            while ((line = br.readLine()) != null) {
                lines++;
                chars += line.length();
                String trimmed = line.trim();
                if (!trimmed.isEmpty()) {
                    words += trimmed.split("\\s+").length;
                }
            }
            br.close();

            System.out.println("Number of lines      : " + lines);
            System.out.println("Number of words      : " + words);
            System.out.println("Number of characters : " + chars);
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:-

```
Number of lines      : 3
Number of words      : 22
Number of characters : 119
```

Q4. Append Data to an Existing File

Create a Java program that appends new student information to an existing file named students.txt without deleting existing contents. Accept Student Name, Roll Number, and Marks from the user and add the record at the end.

students.txt (input file):

```
Rahul, 101, 85
Anjali, 102, 91
```

```
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name : ");
        String name = sc.nextLine();
        System.out.print("Enter Roll Number  : ");
        String roll = sc.nextLine();
        System.out.print("Enter Marks        : ");
        String marks = sc.nextLine();

        try {
            // true = append mode, existing data is not deleted
            FileWriter fw = new FileWriter("students.txt", true);
            fw.write(name + ", " + roll + ", " + marks + "\n");
            fw.close();
            System.out.println("Record appended successfully to students.txt");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
        sc.close();
    }
```

```
}
```

OUTPUT:-

```
Enter Student Name : Aryan
Enter Roll Number  : CU25260022
Enter Marks        : 95
Record appended successfully to students.txt
```

students.txt (after running):

```
Rahul, 101, 85
Anjali, 102, 91
Aryan , CU25260022, 95
```

Q5. Copy One File into Another File

Write a Java program to copy the contents of source.txt into backup.txt. Display a message after the copying operation is completed.

source.txt (input file):

```
This is the source file.
Its content will be copied to backup.txt.
```

```
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

public class Main {
    public static void main(String[] args) {
        try {
            FileReader fr = new FileReader("source.txt");
            FileWriter fw = new FileWriter("backup.txt");

            int ch;
            while ((ch = fr.read()) != -1) {
                fw.write(ch);
            }
            fr.close();
            fw.close();

            System.out.println("File copied successfully from source.txt to backup.txt");
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:-

```
File copied successfully from source.txt to backup.txt
```

backup.txt (after running):

```
This is the source file.
Its content will be copied to backup.txt.
```

Q6. Search for a Word in a File

Create a text file named notes.txt. Write a Java program that accepts a word from the user and searches for that word in the file. Display whether the word is found and the number of times it occurs.

notes.txt (input file):

```
Java is object oriented.
Java is platform independent.
Learning java needs regular practice.
```

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.IOException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the word to search: ");
        String word = sc.nextLine().trim();

        int count = 0;
        try {
            BufferedReader br = new BufferedReader(new FileReader("notes.txt"));
            String line;
            while ((line = br.readLine()) != null) {
                if (line.toLowerCase().contains(word.toLowerCase())) {
                    String[] parts = line.split("\\s+");
```

```
                    for (String w : parts) {
                        // remove punctuation before comparing
                        String clean = w.replaceAll("[^a-zA-Z0-9]", "");
                        if (clean.equalsIgnoreCase(word)) {
                            count++;
                        }
                    }
                }
            }
            br.close();

            if (count > 0) {
                System.out.println("Word \"" + word + "\" found.");
                System.out.println("Number of occurrences: " + count);
            } else {
                System.out.println("Word \"" + word + "\" not found in the file.");
            }
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
        sc.close();
    }
}
```

OUTPUT:-

```
Enter the word to search: java
Word "java" found.
Number of occurrences: 3
```

Q7. Display File Information

Write a Java program that accepts a file name from the user and displays whether the file exists, file name, absolute path, file size, whether it is a file, whether it is a directory, whether it is readable, and whether it is writable.

```
import java.io.File;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter file name: ");
        String name = sc.nextLine();

        File f = new File(name);

        if (f.exists()) {
            System.out.println("File exists      : true");
            System.out.println("File name        : " + f.getName());
            System.out.println("Absolute path    : " + f.getAbsolutePath());
            System.out.println("File size        : " + f.length() + " bytes");
            System.out.println("Is a file        : " + f.isFile());
            System.out.println("Is a directory   : " + f.isDirectory());
            System.out.println("Readable         : " + f.canRead());
            System.out.println("Writable         : " + f.canWrite());
        } else {
            System.out.println("File exists      : false");
            System.out.println("The file \"" + name + "\" does not exist.");
        }
        sc.close();
    }
}
```

OUTPUT:-

```
Enter file name: student.txt
File exists      : true
File name        : student.txt
Absolute path    : /home/claude/w/student.txt
File size        : 85 bytes
Is a file        : true
Is a directory   : false
Readable         : true
Writable         : true
```

Q8. Count Vowels, Consonants, Digits and Spaces

Create a text file named input.txt. Write a Java program to read the file and count the number of vowels, consonants, digits, and spaces. Analyze every character present in the file.

input.txt (input file):

```
Hello World 123 Java 2026
```

```
import java.io.FileReader;
import java.io.IOException;

public class Main {
    public static void main(String[] args) {
        int vowels = 0, consonants = 0, digits = 0, spaces = 0;
        try {
            FileReader fr = new FileReader("input.txt");
            int c;
            while ((c = fr.read()) != -1) {
                char ch = (char) c;
                if (Character.isLetter(ch)) {
                    char l = Character.toLowerCase(ch);
                    if (l == 'a' || l == 'e' || l == 'i' || l == 'o' || l == 'u') {
                        vowels++;
                    } else {
                        consonants++;
                    }
                } else if (Character.isDigit(ch)) {
                    digits++;
                } else if (ch == ' ') {
                    spaces++;
                }
            }
            fr.close();

            System.out.println("Vowels     : " + vowels);
            System.out.println("Consonants : " + consonants);
            System.out.println("Digits     : " + digits);
            System.out.println("Spaces     : " + spaces);
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

OUTPUT:-

```
Vowels     : 5
Consonants : 9
Digits     : 7
Spaces     : 4
```

Q9. Student Record File

Write a Java program to accept details of 5 students and store them in student_records.txt. For each student store Roll Number, Name, Course, and Marks. After storing the records, read the file and display all student records.

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        try {
            FileWriter fw = new FileWriter("student_records.txt");
            for (int i = 1; i <= 5; i++) {
                System.out.println("Enter details of student " + i + ":");
                System.out.print("Roll Number : ");
                String roll = sc.nextLine();
                System.out.print("Name        : ");
                String name = sc.nextLine();
                System.out.print("Course      : ");
                String course = sc.nextLine();
                System.out.print("Marks       : ");
                String marks = sc.nextLine();
                fw.write(roll + "," + name + "," + course + "," + marks + "\n");
            }
            fw.close();
            System.out.println("\nRecords stored in student_records.txt");

            BufferedReader br = new BufferedReader(new FileReader("student_records.txt"));
            String line;
            System.out.println("\n----- Student Records -----");
            System.out.printf("%-12s %-12s %-8s %-6s%n", "Roll No", "Name", "Course", "Marks");
            while ((line = br.readLine()) != null) {
                String[] p = line.split(",");
                System.out.printf("%-12s %-12s %-8s %-6s%n", p[0], p[1], p[2], p[3]);
            }
            br.close();
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
        sc.close();
    }
}
```

OUTPUT:-

```
Enter details of student 1:
Roll Number : 101
Name        : Aman
Course      : BCA
Marks       : 88
Enter details of student 2:
Roll Number : 102
Name        : Riya
Course      : BCA
Marks       : 92
Enter details of student 3:
Roll Number : 103
Name        : Karan
Course      : BCA
Marks       : 79
Enter details of student 4:
Roll Number : 104
Name        : Sneha
Course      : BCA
Marks       : 95
Enter details of student 5:
Roll Number : 105
Name        : Aryan
Course      : BCA
Marks       : 97

Records stored in student_records.txt

----- Student Records -----
Roll No      Name         Course   Marks
101          Aman         BCA      88
102          Riya         BCA      92
103          Karan        BCA      79
104          Sneha        BCA      95
105          Aryan        BCA      97
```

Q10. File-Based Employee Management

Create a file-based employee management program. Accept Employee ID, Employee Name, Department, and Salary and store them in employees.txt. Then read the file and search for an employee using Employee ID. If found, display the complete record; otherwise display 'Employee record not found.' Handle the situation where the file does not exist.

```
import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        File file = new File("employees.txt");

        try {
            System.out.print("Enter number of employees: ");
            int n = Integer.parseInt(sc.nextLine());

            FileWriter fw = new FileWriter(file);
            for (int i = 1; i <= n; i++) {
                System.out.println("Enter details of employee " + i + ":");
                System.out.print("Employee ID   : ");
                String id = sc.nextLine();
                System.out.print("Employee Name : ");
                String name = sc.nextLine();
                System.out.print("Department    : ");
                String dept = sc.nextLine();
                System.out.print("Salary        : ");
                String sal = sc.nextLine();
                fw.write(id + "," + name + "," + dept + "," + sal + "\n");
            }
            fw.close();
            System.out.println("Records saved in employees.txt\n");
        } catch (IOException e) {
            System.out.println("Error while writing file: " + e.getMessage());
        } catch (NumberFormatException e) {
            System.out.println("Invalid number entered.");
            sc.close();
            return;
        }

        System.out.print("Enter Employee ID to search: ");
        String searchId = sc.nextLine().trim();

        if (!file.exists()) {
            System.out.println("Error: employees.txt does not exist.");
            sc.close();
            return;
        }

        boolean found = false;
        try {
            BufferedReader br = new BufferedReader(new FileReader(file));
            String line;
            while ((line = br.readLine()) != null) {
                String[] p = line.split(",");
                if (p[0].equals(searchId)) {
                    System.out.println("\nEmployee record found:");
                    System.out.println("Employee ID   : " + p[0]);
                    System.out.println("Employee Name : " + p[1]);
                    System.out.println("Department    : " + p[2]);
                    System.out.println("Salary        : " + p[3]);
                    found = true;
                    break;
                }
            }
            br.close();
        } catch (IOException e) {
            System.out.println("Error while reading file: " + e.getMessage());
        }

        if (!found) {
            System.out.println("Employee record not found.");
        }
```

```
        sc.close();
    }
}
```

OUTPUT:-

```
Enter number of employees: 3
Enter details of employee 1:
Employee ID   : E101
Employee Name : Aman
Department    : Sales
Salary        : 45000
Enter details of employee 2:
Employee ID   : E102
Employee Name : Riya
Department    : HR
Salary        : 52000
Enter details of employee 3:
Employee ID   : E103
Employee Name : Karan
Department    : IT
Salary        : 60000
Records saved in employees.txt

Enter Employee ID to search: E102

Employee record found:
Employee ID   : E102
Employee Name : Riya
Department    : HR
Salary        : 52000
```